Preliminary Shielding Assessment - New build

A client-facing preliminary assessment for design development and shielding coordination. It identifies the adopted design basis, preliminary barrier requirements, and the information that must be confirmed before final shielding acceptance.

This report is not a certificate of shielding compliance.

1. Project details

Project name	NCRP147 Section 5.3 - UI approximation (NOT strict KAT)
Room name	X-ray Room 1
Modality	X-RAY
Project type	New build
Project note	IMPORTANT: Not an exact NCRP 147 Section 5.3 fixture. UI fixed south-facing table source cannot reproduce cross-table lateral primary beam.
Client	Physics validation - reference test
Site	NCRP 147 Section 5.3 representative room
Assessor	Health Physics Pty Ltd
Date	2026-10-14
Report status	Draft
Revision	Rev 0

2. Assessment outcome

Overall preliminary outcome	**FAIL**
Barriers assessed	9
Barriers requiring review / redesign	4
Design decision	Do not rely on the current shielding arrangement for construction; redesign the failed barrier(s) and reassess.

3. Design basis

The preliminary result is valid only for the room geometry, source/workload and occupancy assumptions summarised below. These values should be confirmed by the client/design team before detailed design.

3.1 Room and geometry

Room	X-ray Room 1
Modality	X-ray
Room dimensions	6.50 m x 4.00 m
FFL to FCL	3.60 m
Source position	4.20 m from west | 1.20 m from south

3.2 Source and workload assumptions

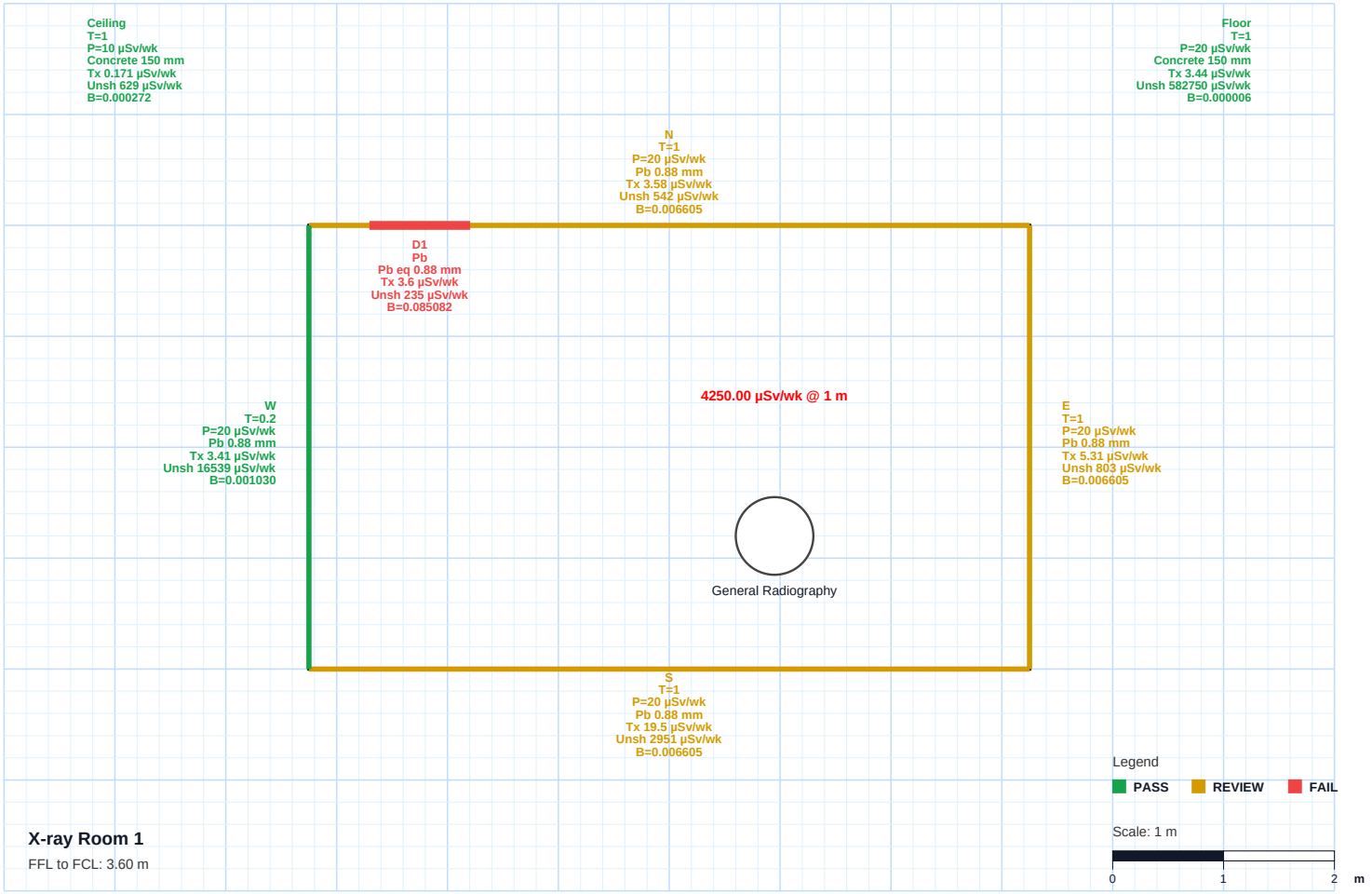
Parameter	Adopted preliminary input
Source	General Radiography
Type / radiation	X-ray primary | xray 120kvp
Shielding method	NCRP 147 simplified general radiography
Exam room type	General Radiography
Examinations per week	125 exams/week
Tube potential	120 kVp

3.3 Design criteria and occupancy

Barrier	Adjacent area	Distance (m)	Occupancy T	P (uSv/wk)
N wall	Corridor	2.80	1.000	20.0
E wall	Corridor	2.30	1.000	20.0
S wall	Corridor	1.20	1.000	20.0
W wall	Corridor	4.20	0.200	20.0
Floor	Clinical Area	1.00	1.000	20.0

Barrier	Adjacent area	Distance (m)	Occupancy T	P (uSv/wk)
Ceiling	Clinical Area	2.60	1.000	10.0
Control wall	Control booth	1.72	1.000	100.0
W Bucky shield	Corridor	4.20	0.200	20.0
D1	N wall	4.25	1.000	20.0

4. Barrier plan



Barrier labels show the entered construction and the calculated preliminary result. The architectural drawing, openings and service penetrations must be coordinated separately before final acceptance.

5. Barrier schedule

The schedule below reports the canonical calculation results for the assessed barriers.

Barrier	Actual (mm)	Role	Distance (m)	T	P (uSv/wk)	Unshielded (uSv/wk)	Transmitted (uSv/wk)	B	Required (mm)	Solver	Status
N wall	0.88	Fig52 Secondary Wall	2.80	1.000	20.000	542.092	3.580	0.006605	0.41	ncrp147_s5.3_fig52_secondary_wall	REVIEW
E wall	0.88	Fig52 Secondary Wall	2.30	1.000	20.000	803.403	5.306	0.006605	0.50	ncrp147_s5.3_fig52_secondary_wall	REVIEW
S wall	0.88	Fig52 Secondary Wall	1.20	1.000	20.000	2951.389	19.493	0.006605	0.87	ncrp147_s5.3_fig52_secondary_wall	REVIEW
W wall	0.88	Fig52 Chest Bucky Wall	4.20	0.200	20.000	16539.116	3.406	0.001030	0.12	ncrp147_s5.3_fig52_chest_bucky_wall	PASS
Floor	150.00	Fig52 Floor Under Table	1.00	1.000	20.000	582750.000	3.441	0.000006	97.30	ncrp147_s5.3_fig52_floor_under_table	PASS
Ceiling	150.00	Fig52 Ceiling	2.60	1.000	10.000	628.698	0.171	0.000272	51.70	ncrp147_s5.3_fig52_ceiling	PASS
Control wall	0.88	Fig52 Control Wall	1.72	1.000	100.000	1439.458	9.507	0.006605	0.28	ncrp147_s5.3_fig52_control_wall	PASS
W Bucky shield	2.20	Fig52 Chest Bucky Subbarrier	4.20	0.200	20.000	4.228	0.001	0.001529		ncrp147_s5.3_fig52_chest_bucky_wall	PASS
D1	0.88	Door	4.25	1.000	20.000	235.066	3.596	0.085082	1.00	mu_plus_buildup	FAIL

Chest Bucky Shield primary attenuation credit:

W Bucky shield: Finite 1.2 m Chest Bucky sub-barrier: all radiation traverses structural wall plus Pb 2.20 mm Bucky Shield; primary Bbs=0.000248065.

6. Design actions and verification

This page identifies the items that should be closed out before the preliminary assessment is relied upon for construction or final shielding acceptance.

6.1 Project coordination actions

Action	Suggested responsibility	Required before
Confirm source workload, occupancy classification and installed barrier thickness.	Client / design team / physicist	Design freeze
Confirm source workload, occupancy classification and installed barrier thickness.	Client / design team / physicist	Design freeze
Confirm source workload, occupancy classification and installed barrier thickness.	Client / design team / physicist	Design freeze
Increase lead shielding to at least 1.00 mm and confirm installed thickness.	Client / design team / physicist	Design freeze
Confirm the current architectural drawing number/revision, room orientation and dimensions.	Architect / client	Detailed assessment
Confirm the use and occupancy of areas beyond each wall, floor and ceiling.	Client / architect	Detailed assessment
Provide door, viewing panel, duct, conduit and service-penetration details so shielding continuity can be assessed.	Architect / builder	Construction documentation
Confirm radionuclide/scanner workload and operating assumptions with the intended operator.	Imaging operator / client	Detailed assessment

6.2 Changes that trigger reassessment

- Any change to room dimensions, source position, adjacent-area use or occupancy.
- Any change to barrier construction, doors, glazing, penetrations or shielding continuity.
- Addition of radiation-producing equipment or a source not represented in this assessment.

7. Conclusion and next stage

Overall preliminary outcome	**FAIL**
Barrier result	5 PASS | 3 REVIEW | 1 FAIL
Client decision	Do not rely on the current shielding arrangement for construction; redesign the failed barrier(s) and reassess.

7.1 Preliminary conclusion

The preliminary assessment contains 1 barrier failure(s). The affected barriers require redesign and reassessment before the shielding arrangement is accepted for construction.

7.2 Recommended next stage

For a new build, the next stage should coordinate the shielding schedule with architectural drawings, door/window schedules, penetrations and confirmed operating assumptions before issuing a detailed/construction shielding assessment.

7.3 Information normally required for the detailed/final assessment

- Current coordinated architectural drawings showing room dimensions, wall construction, doors, glazing and service penetrations.
- Confirmed use and occupancy of all adjacent areas, including above and below the room.
- Confirmed clinical workload/source assumptions and any additional radiation-producing equipment.

8. Preliminary assessment limitations

- This document is a preliminary shielding assessment for design development and shielding review; it is not a final shielding compliance certificate.
- Final shielding acceptance requires professional review of architectural drawings, construction details, source and workload assumptions, occupancy assumptions, and installed materials.
- Calculated results depend on the entered room geometry, source position, operating assumptions, design goals, occupancy factors, and barrier construction.
- General radiography results use the NCRP Report No. 147 Section 5.3 / Figure 5.2 radiographic-room methodology with project-specific barrier distances.
- Primary-beam pre-shielding is explicit for the floor, chest-bucky and cross-table scenarios; cross-table lateral shielding combines transmitted primary and secondary air kerma iteratively.
- Changes to examination workload, tube potential, source or wall-bucky position, primary-beam use, occupancy, design goals or barrier construction require reassessment.

• Issueable Preliminary Shielding Assessment output is limited to the qualified NCRP 147 General Radiography pathway; unsupported or manual X-ray configurations require separate assessment.